Eduardo Escoto

Second Year CS PhD Student at UC San Diego, NSF CSGrad4US Fellow

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Objective

I am a second-year Ph.D. student funded by the National Science Foundation, working under the guidance of Dr. Julian McAuley. My current research interests are in the foundations of Representation Learning (Autoencoders, Vector Quantization) and Generative Models (Diffusion, Flow Models). Additionally, I am interested in their applications, right now I am particularly working on applications in fields like Recommendations, Information Retrieval, and Music.

Education

University of California, San Diego *San Diego, CA*
Doctor of Philisophy in Computer Science *Sept 2024 – June 2029*

- **Advisor**: Julian McAuley
- **Funding and Fellowships**: First Year Departmental Fellowship, NSF CSGrad4US Fellow
- **Research**: Generative Models, Recommender Systems, Retrieval Systems, Representation Learning

University of California, Santa Barbara *Santa Barbara, CA*
Bachelor of Science (Double Major) in Mathematics and Statistics *Sept 2017 – June 2020*

- **Concentration**: Probability and Statistics
- **Organizations**: Data Science Club, Association for Women in Mathematics
- **Relevant Coursework**: Bayesian Statistics, Data Science, Machine Learning, Stochastic Systems, Real Analysis, Numerical Analysis, Complex Analysis, Graph Theory, Abstract/Linear Algebra

San Diego Miramar College *San Diego, CA*
Assosciate of Arts in Mathematics *Sept 2014 – June 2017*

- **Accomplishments**: Honors Student, Dean's List Scholar, Annual Research Symposium Presenter
- **Organizations**: Phi Theta Kappa Honors Society
- **Relevant Coursework**: Calculus I/II/III, Discrete Mathematics, C++, Java, Visual Basic

Experience

Machine Learning Engineer, PhD Intern *San Francisco, CA*
Instacart *June 2025 – Sept 2026*

This was a 16 Week internship, focused on research and production implementation of a Semantic ID based recommender and search system.

- **Semantic ID and Neural Representations**: Built a comprehensive Semantic ID system to improve various systems across Instacart's infrastructure. Applications that it has improved range from generative recommendations, catalog and taxonomy, and search.
- **Efficient Visual / Multimodal Search**: A direct application of my Semantic ID system has been creating an efficient visual search, and Multimodal search procedure. Patents have been filed and are currently pending.

Senior Data Scientist *New York, NY*

Noom *Jan 2022 – Sept 2024*

This was a Product Data Scientist role mainly focused on A/B testing on Growth, Revenue optimization, and the Acquisition Funnel.

- **Cross-Functional Partnership**: Coordinated collaboration between Data Science, Data Engineering, and Engineering teams to ensure successful implementation of data requirements and infrastructure for event-based data consumption and model deployment.
- **A/B Testing and Experiment Design**: Aided in design and development of A/B Tests and Experiments for Growth. Analyzed experiments results and helped stakeholders in pinpointing findings and high value results.
- **ETL and BI Development**: Developed Spark and Redshift ETL pipelines for data analysis and for use across the organization. Communicated and documented data requirements from stakeholders and other Data Scientists. Built LookML code for Looker Dashboards and Explores to use new data sources.
- **User Behavior Analysis**: Incorporated techniques like Causal Inference, Clustering, and Machine Learning to utilize observational and experimental data in order to understand user behavior and to inform product decisions.
- **Machine Learning**: Built ML models for Real-Time Personalization, Recommendation, and NLP from free text entry data. Leveraged LLM's for building AI driven products.

Data Scientist *Washington, DC*

Afiniti *Aug 2020 – Dec 2021*

This was a Machine Learning Data Scientist role focused on building Bayesian models for behavior based pairings of agents and clients.

- **Built Traditional and Deep Learning Models**: Prototyped, engineered, and pipelined a plethora of NLP models, embedding models, and predictive models. Utilized traditional ML methods like Gradient Boosted Decision Trees and Random Forests for models with tabular data.
- **Bayesian Data Analysis & Modeling**: Created STAN models to model interaction between call center agents and callers. Researched and improved models through iteration and testing.
- **Code Contribution and Numerical Computing**: Maintained company code repositories for Machine Learning and Modeling. Implemented and optimized new features in our modeling package.

Data Science Intern *Santa Barbara, CA*

Dept. of Health & Wellness | University of California, Santa Barbara *May 2019 – June 2020*

- **Facilitated Data Gathering and Management**: Leveraged platforms like Qualtics and Microsoft's Power BI in order to centralize data gathering and availability. Monitored survey question quality, and response quality.
- **Analyzed Data and Builds Models**: Incorporated a wide array of tools and techniques to analyze all departmental and survey data. These included Bayesian Methods, Machine Learning, and Classical Methods.
- **Optimized Internal Processes**: Innovated departmental systems and replaced them with modern analogues. Built web–based solution to streamline inventory and key management with a custom React-GraphQL App.

Software Engineering Intern *San Diego, CA*

Biosero Inc. *Sept 2016 – Sept 2017*

- **Database Engineering and Driver Development**: Engineered and optimized microplate tracking database. Developed drivers for a myriad of Bio-tech instruments for use in automization. These included Robotic Arms Sealers/Desealers, Acoustic Dispensing Systems, Microplate Readers, and many more.
- **MagneMotion Track Integration**: Implemented magnetic track system for use in large scale and connected integrations. Utilized graph theory and queueing theory to streamline track movement algorithm.

Barista & Shift Supervisor *San Diego, CA*

Starbucks Coffee *June 2013 – June 2019*

- **Coffee and Food Master**: Succeeded in Coffee and Food master certifications and continued on to win the district Barista Championship in 2016. First Partner of the Quarter at store.
- **Established Leadership and Teamwork Skills**: Led a team of world-class baristas to create a positive, inclusive, and motivating work environment that customers consistently enjoy visiting.

Teaching Experience

Undergraduate Learning Assistant *Santa Barbara, CA*
Dept. of Computer Science | University of California, Santa Barbara *Mar 2019 – Dec 2020*

- **INT 5**: Data Science Foundations
- **INT 15**: Data Science Tools and Techniques
- **Created Curriculum**: Worked with team to meticulously write guided labs and homeworks in Notebooks. Created resources for students to use in minimizing programming learning curve.
- **Engineered and Maintained Autograding Tooling**: Developed python scripts in order to autograde Jupyter Notebook assignments and upload scores and automatic feedback to Grading Portal via API.

Publications

When Data Scaling isn't Enough: Leveraging Video Scene Graphs to Improve Compositional Reasoning in Video-Language Models Feb 2026
Kibum Kim, Kyle Min, Eduardo Escoto, Yueqi Wang, Julian McAuley, Chanyoung Park
openreview.net/forum?id=pxB1lmkVAD (ICML 2026 - Submission Awaiting Review)

Expressiveness Limits of Autoregressive Semantic ID Generation in Generative Recommendation Feb 2026
Yupeng Hou, Haven Kim, Mingxuan Ju, Eduardo Escoto, Neil Shah, Julian McAuley
openreview.net/forum?id=oixY2dwih6 (ICML 2026 - Submission Awaiting Review)

Honors

NSF CSGrad4US Fellowship: Fully funded by the NSF for 3 years of my PhD. The fellowship's goal is to increase the number of domestic graduate students pursuing research and innovation careers in Computer Science.

Skills

Languages: Python, SQL, Rust, C++, Julia, Haskell, Javascript, Shell Scripting

ML Frameworks: PyTorch, JAX, TensorFlow, Scikit-Learn, XGBoost, LightGBM, Lightning, HuggingFace

Infrastructure: Distributed training, AWS, SLURM, Lightning Fabric

Research Areas: Generative Models, Diffusion, Flow Matching, Semantic IDs, Representation Learning, Recommendation, Retrieval, Multi-Modal Models